NLP for Teens

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▪ TF-IDF

TF-IDF stands for Term Frequency – Inverse Document Frequency.
It’s a way to find important words in a document.

🔹 TF = How often a word appears in ONE document
If a word appears many times in a document → it might be important.

🔹 IDF = How rare the word is across ALL documents
If a word appears in every document → it’s not special.
If a word appears in only one document → it’s special.

- TF-IDF

- TF-IDF = TF × IDF

So a word gets a high score when:

It appears many times in one document (high TF)

It appears rarely in other documents (high IDF)

This helps computers understand which words matter.

▪ Why TF-IDF is useful

It removes boring words like the, is, and, good
It highlights meaningful words like battery, refund, delivery
It helps in search engines, chatbots, spam filters, and recommendations

▪ TF-IDF Example

Imagine three reviews:

“The food is good”

“The service is good”

“Good food and good service”

“good” appears everywhere → not special

“food” appears in fewer documents → more special

“service” appears in fewer documents → more special

TF-IDF will give higher scores to “food” and “service”.

Assignment

- This assignment uses the SMS Spam dataset to apply TF-IDF, run Stratified K-Fold, build pipelines, train multiple models, compare their accuracy, and report precision/recall/F1 for spam detection.

Summary

- ✓ TF-IDF — converting text into weighted word features
- ✓ Stratified K-Fold — balanced cross-validation splits
- ✓ Pipeline — TF-IDF + model combined
- ✓ Model Comparison — evaluating KNN, Logistic Regression, Random Forest
- ✓ Classification Report — precision, recall, F1 for spam detection

Thank You

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